

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT (open to sponsorship), open to SF relocation

PROFESSIONAL SUMMARY

Forward Deployed AI Engineer with production agentic LLM systems and ERP-grade cross-system integration experience. Built and shipped a 5-stage multi-agent healthcare claims pipeline with schema-validated JSON contracts between agents - the reliability discipline embedded enterprise AI deployments need to prevent cascading hallucinations. Owned Order-to-Cash ERP modules (contracts, nominations, allocations, invoicing) at Emerson, functionally analogous to NetSuite/Salesforce data-flow mapping. Packaged production AI as FastAPI/Flask services in Docker (~30% post-deploy defect reduction). Healthcare RAG with ~35% retrieval precision gain and >90% grounding alignment. Master's in CS (Kent State). US-based, F-1 OPT, open to relocating to San Francisco.

CORE COMPETENCIES

Production AI Agents • Exception Routing & Cross-System Coordination • Data Reconciliation • ERP Integration Patterns • Multi-Agent Orchestration • Schema-Validated Contracts • RAG & Eval Harnesses • Python • TypeScript • FastAPI • Docker • PostgreSQL / T-SQL

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Healthcare AI, Agentic Workflows, Production Delivery

- Built **agentic LLM workflows** for multi-step healthcare queries (eligibility, care workflow navigation, documentation) with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~25% - the same reliability pattern enterprise AI agents need.
- Shipped **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved contextual retrieval precision ~35% and reduced irrelevant retrieval >30%. Grounding alignment exceeded 90% in evaluation.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects ~30%.
- Delivered HIPAA-conscious system-limitation docs and stakeholder-facing audit trails - the documentation pattern Fortune 2000 client engagements depend on.
- Shipped predictive ML pipelines (scikit-learn, XGBoost); improved recall 15-20% on high-risk categories while holding precision >90%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India

Database & DevOps Performance Engineer (Intern) - ERP Cross-System Integration, Compliance-Sensitive Oil & Gas

- Optimized **Synthesis Order-to-Cash ERP platform** (contracts, nominations, allocations, invoicing) - functionally analogous to NetSuite/Salesforce cross-system data flow mapping; reduced query execution ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines (Jenkins)** for version-controlled SQL and stored-procedure deploys with structured release validation and rollback checkpoints; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed **SQL Server DMV + Grafana** performance dashboards to track cross-system long-running queries, deadlocks, and CPU/I/O contention; cut incident recurrence ~25% and improved RCA speed ~30%.
- Tuned reconciliation queries (index rebuilds, statistics updates, partition-aware indexing); improved reconciliation job runtime ~15% - the same reconciliation pattern enterprise AI agents automate.
- Supported **RBAC + audit logging** for financial modules in a compliance-sensitive oil & gas environment.

Machine Learning Engineer - Applied NLP, Document Q&A with RAG

- Built a transformer-based summarization system over **5,000+ sessions**; hierarchical long-context summarization cut manual review time **60-70%**.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%**, raised answer accuracy **>85%**.
- Deployed the stack as a **Flask** API with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Pipeline - Schema-Contracted Multi-Agent System

Multi-agent pipeline (Intake Normalization -> Validation -> Consistency Analysis -> Duplicate Detection -> Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy docs, ANN duplicate detection, explainable risk scoring with **audit-ready reasoning traces**. The same multi-agent reliability discipline embedded enterprise AI deployments need.

Multi-agent orchestration, JSON schema contracts, RAG, vector search, audit traces

Manga Lens - Chrome Extension Integrating 4 AI Vision Providers (shipped, Chrome Web Store)

Full-stack browser extension built independently. Manifest V3, content scripts, service workers, captureVisibleTab for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) behind one interface with **provider-specific payload handling**. Seven-day translation cache, per-domain selector configs for 29 sites, narrowed host permissions.

TypeScript, Manifest V3, multi-provider AI integration, payload abstraction

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs (Perplexity Sonar + Replicate). **Intermediate structured prompt transformation layers** improved contextual alignment **~30%** and first-pass image success **~25-30%**.

FastAPI, TypeScript, structured prompt orchestration

Driver Drowsiness Detection - Real-Time YOLOv8 with Adaptive Throttling

Replaced a two-stage CNN pipeline with a **unified YOLOv8** model; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives (**~25% reduction**).

YOLOv8, OpenCV, real-time CV, latency engineering

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite), React, Node.js

LLM & GenAI: Production AI agents, RAG, agentic workflows, exception routing, cross-system coordination, schema contracts, structured outputs, eval pipelines, grounding, embeddings, vector search

Infra & DevOps: Docker, Jenkins, CI/CD, RESTful APIs, observability/logging, Grafana, SQL Server DMVs

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Enterprise / Compliance: ERP integration patterns, HIPAA-conscious governance, RBAC, audit logging, release validation, rollback checkpoints